

FIRST ARTICLE INSPECTION (FAI) DECISION & AUDIT DASHBOARD

Document Reference: FAI-DEC-2026-012 | Executive Quality Audit Clearance

Supplier Name: AeroMotion Dynamics Ltd.	Part Name: Brushless Motor Housing
Part Number: AMD-EV-7042	Application: EV & Drone Drive System
Inspection Date: August 2026	Final Status: HIGH RISK (FAI REJECTED)

1. EXECUTIVE SUMMARY & AUDIT DECISION

This document serves as the official First Article Inspection (FAI) Decision Certificate issued under the authority of the Head of Operations. Following rigorous automated sensor scanning and ISO compliance auditing, the initial production batch of **Brushless Motor Housing (Part # AMD-EV-7042)** submitted by AeroMotion Dynamics Ltd. has been **REJECTED** due to critical quality non-conformances.

2. DUAL-LAYER AUTOMATED SENSOR SCAN EVALUATION

To maintain zero-defect standards across critical aerospace and electric vehicle drive applications, the component batch underwent two-tier automated sensor evaluation:

• Layer 1: Operational & Dimensional Sensor Scan — [FAILED]

Dimensional inspection sensors detected severe out-of-tolerance conditions exceeding Permissible Limits (0.05mm deviation on key bearing seat diameters), accompanied by localized surface roughness defects.

• Layer 2: ISO Smart Compliance Scanner — [FAILED]

Automated multi-standard scanning flagged critical non-conformances against five core international management and technical standards:

- **ISO 9001:2015:** Failure in shop-floor process control and quality assurance protocols.
- **ISO 14001:2015:** Non-compliance in environmental waste and coolant disposal management.
- **ISO 19011:2018:** Audit trail gaps and missing operational verification documentation.
- **ISO 45001:2018:** Breach of shop-floor operator safety protocols during batch production.
- **ISO 10474 / EN 10204 3.1:** Raw material composition and testing certificate verification failure.

3. PARETO-DRIVEN ROOT CAUSE ANALYSIS & 30-DAY SCAR RECOVERY PLAN

In accordance with Day 11 SOP protocol, a mandatory 30-Day Supplier Corrective Action Request (SCAR) timeline is enforced. The top three root causes identified via Pareto Analysis must be systematically rectified in three 10-day phases:

30-Day Execution & Daily Logging Matrix:

- **Phase 1 (Days 01–10) — Raw Material Quality Failure:** Rectify raw material sourcing, replace non-compliant alloy stock, and secure certified ISO 10474 / EN 10204 3.1 material test reports.
- **Phase 2 (Days 11–20) — Dimensional & Operational Errors:** Recalibrate CNC machinery, eliminate geometric deviations, and enforce Statistical Process Control (SPC) for tight tolerance holding.
- **Phase 3 (Days 21–30) — ISO & EHS Standard Non-Compliance:** Re-align shop-floor operations with ISO 9001, ISO 14001, ISO 19011, and ISO 45001 standards; submit completed daily logs for final FAI resubmission.

4. HEAD OF OPERATIONS GUIDANCE & APPROVAL AUTHORITY (TOYOTA-STYLE MENTORSHIP)

Authority: Head of Operations

Technical Justification & Actionable Feedback:

Rejecting an FAI is a measure to safeguard product integrity and end-user safety in critical drive systems. In line with Toyota-style supplier development philosophy, rejection is accompanied by direct engineering support rather than punitive disconnection:

- **Material Procurement:** Source raw materials exclusively from ISO-certified tier-1 suppliers and attach 3.1 Inspection Certificates with every batch.
- **Process Control:** Implement automated in-line probing on CNC machines to prevent dimensional drift before batch completion.
- **EHS & Audit Alignment:** Retain shop-floor technicians on ISO 45001 safety and ISO 14001 environmental protocols. Complete a internal audit as per ISO 19011 guidelines prior to resubmission.

Authorized Sign-off: Head of Operations | Quality Engineering Division
Status: Conditional Rejection — Enforced 30-Day SCAR Active

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